A Flexible Framework for Harmonizing ICU Waveform and Clinical Data from MIMIC-III

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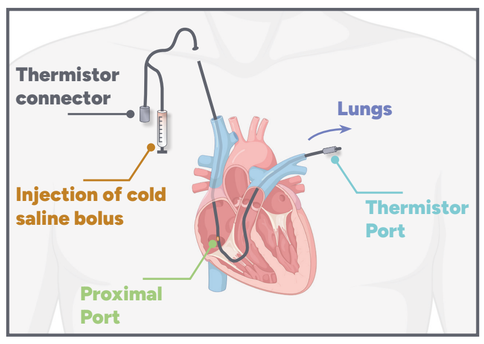
1. Background & Motivation

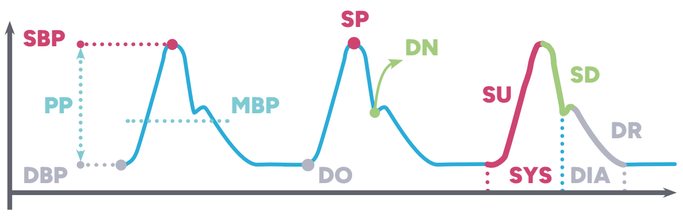
Cardiac Output (CO) Measurement

→ Monitors hemodynamic changes and tissue perfusion
→ Critical in sepsis, shock, or during drug administration

Gold standard in Intensive Care Unit (ICU)

→ Pulmonary Artery Catheter (Swan–Ganz)
→ Thermodilution (TCO): Cold saline bolus injection estimates the blood flow using Stewart-Hamilton law; infection-prone





DBP = DIASTOLIC BLOOD PRESSURE
DIA = DIASTOLE
DO = DIASTOLIC ONSET
DR = DIASTOLIC RUNOFF
PP = PULSE PRESSURE
MBP = MEAN BLOOD PRESSURE
SBP = SYSTOLIC BLOOD PRESSURE
SP = SYSTOLIC PEAK
SU = SYSTOLIC UPSTROKE
SYS = SYSTOLE
SD = SYSTOLIC DECLINE
DN = DICROTIC NOTCH

Arterial Blood Pressure (ABP) Measurement

→ Continuous monitoring of blood pressure
→ ABP waveform morphology varies with:
- Hemodynamic conditions
- Vessel stiffness
- Underlying pathologies
- Drug administration
- Transducer calibration
- Signal artifacts

Gold standard in ICU

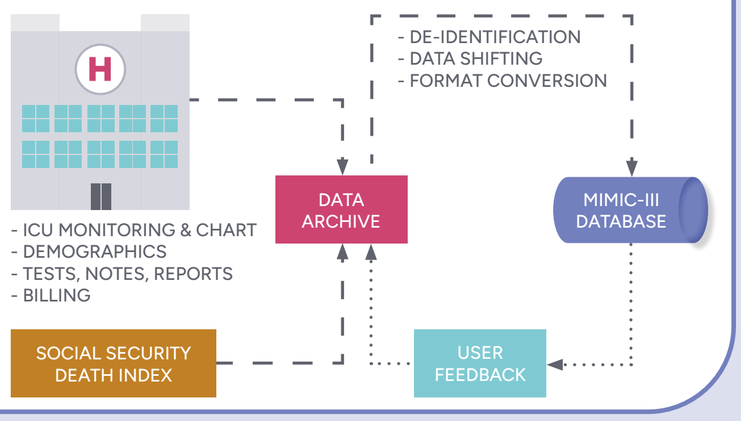
→ Arterial line + fluid-filled strain-gauge

Goals and Motivation

→ Build a dataset linking less invasive ABP signals with CO measurements
→ Use it to train and validate machine learning models for CO prediction

In vivo Data

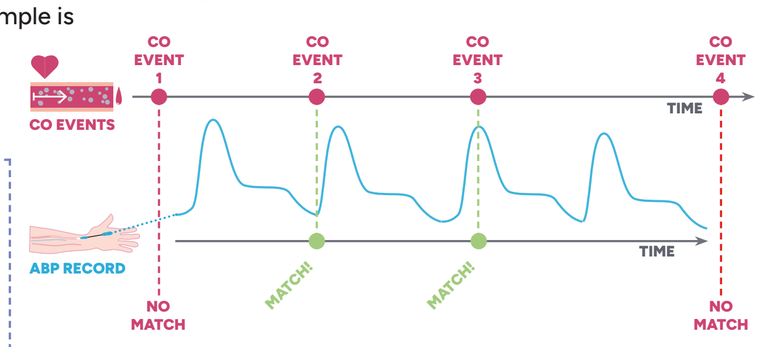
→ ICU *in vivo* data are required for realistic CO prediction
→ MIMIC-III provides extensive ICU records, but a linked ABP-CO dataset with patient information must be built



2. Pipeline Concept

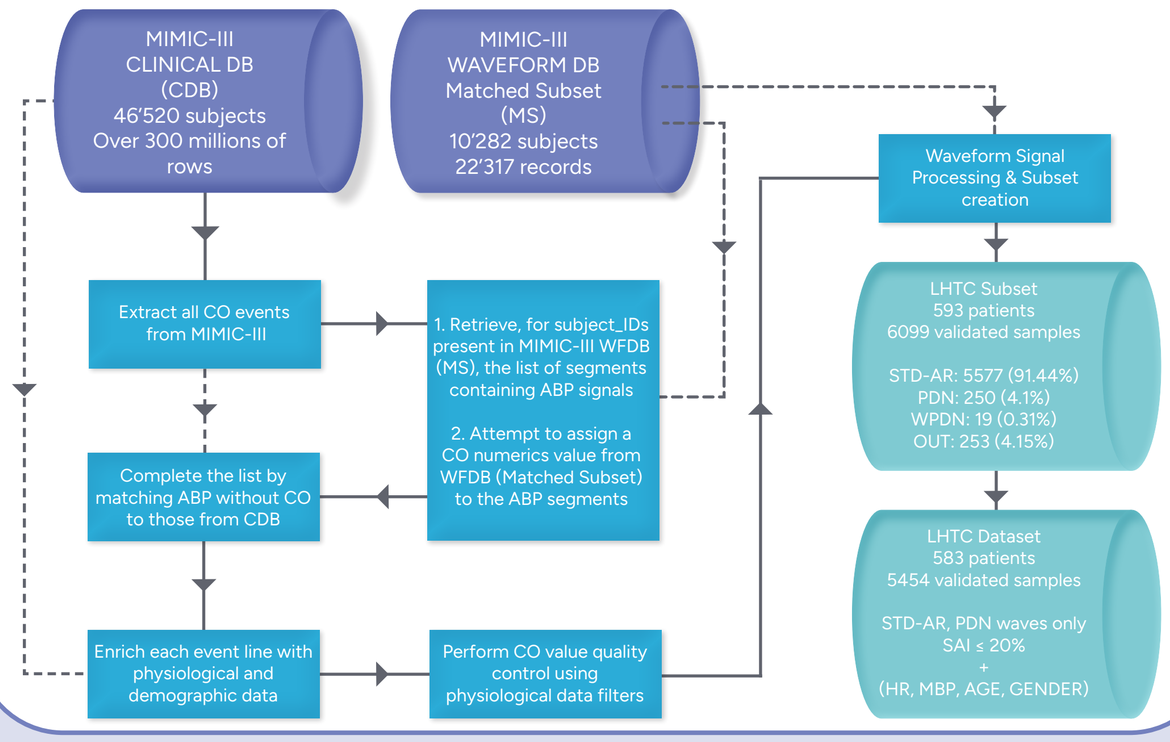
Sample temporal matching

A sample contains a primary feature (ABP wave) and a target (CO) from the same patient. If a CO event occures within a waveform record, a sample is created.
→ Other CO events are discarded
→ Additional data (e.g., demographic) are linked to each sample

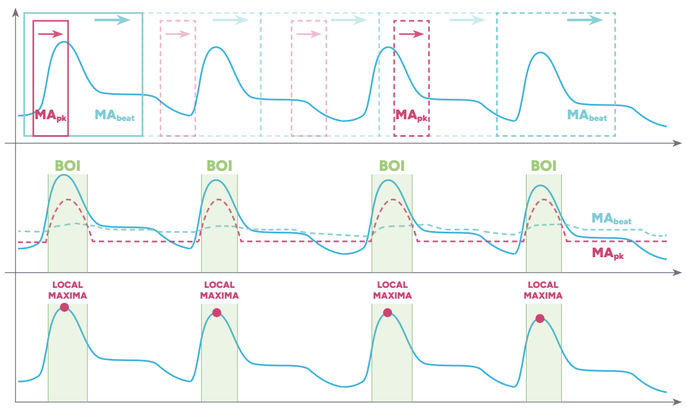


Data retrieval pipeline

→ CO events are present in both the Clinical DB and the Waveform DB
→ Demographics only available in CDB
→ ABP signals only available in the Waveform DB



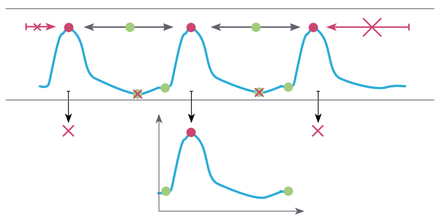
3. Signal Processing Algorithms



Wave isolation

← A dual moving-average filter with adaptive threshold detects systolic peaks

Onsets are located between peaks, allowing waves to be isolated from the segment ↓



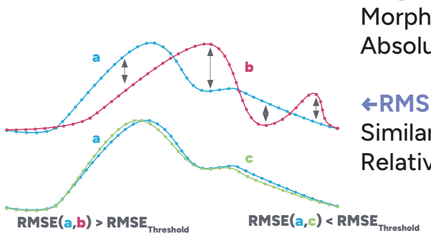
Wave Analysis Algorithms

→ After isolation, three algorithms identify the most representative (dominant) wave for each ABP segment. They reject non-representative or noisy waves and establish a qualitative ranking of candidate waves for dominance based on a multi-criteria approach.

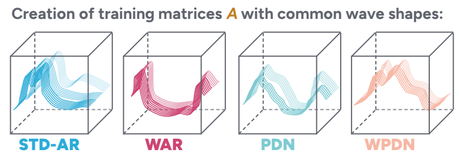
←Signal Abnormality Index
Value & Trend based

Singular Value Decomposition→
Morphological trend
Absolute reference

←RMSE comparison
Similarity based
Relative reference (Segment)



Creation of training matrices *A* with common wave shapes:



Singular Value Decomposition (SVD): $A = USV^T$

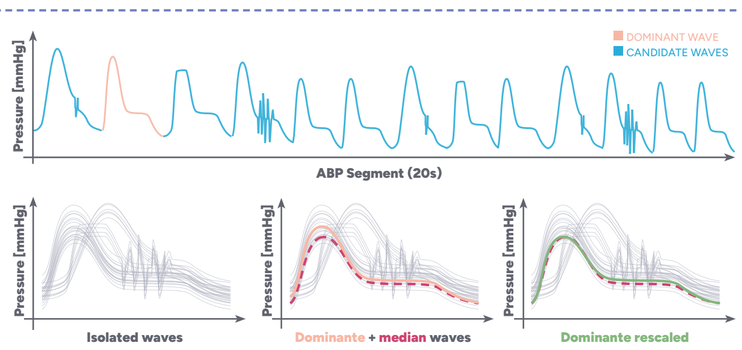
The first *k* component of *U* → Signal subspace

Project new beat into signal subspace → $P_k(ABP) = U_k (U_k^T ABP)$

$$SNR = \frac{\text{Projected signal energy}}{\text{Residual energy}} = \frac{\| P_k(ABP) \|^2}{\| ABP - P_k(ABP) \|^2}$$

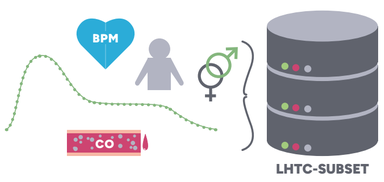
Further Processing

The dominant wave is then rescaled to match the median pressure magnitude of other candidates, and additional filters are applied (e.g., removal of systolic plateau artifacts, exclusion of signals with SAI ≥ 20%, or undesired morphologies).



LHTC-Subset creation

→ All relevant information (raw data, pipeline parameters, results) is gathered into a local subset
→ The algorithm automates sample generation, drastically reducing the manual review time required by human operators



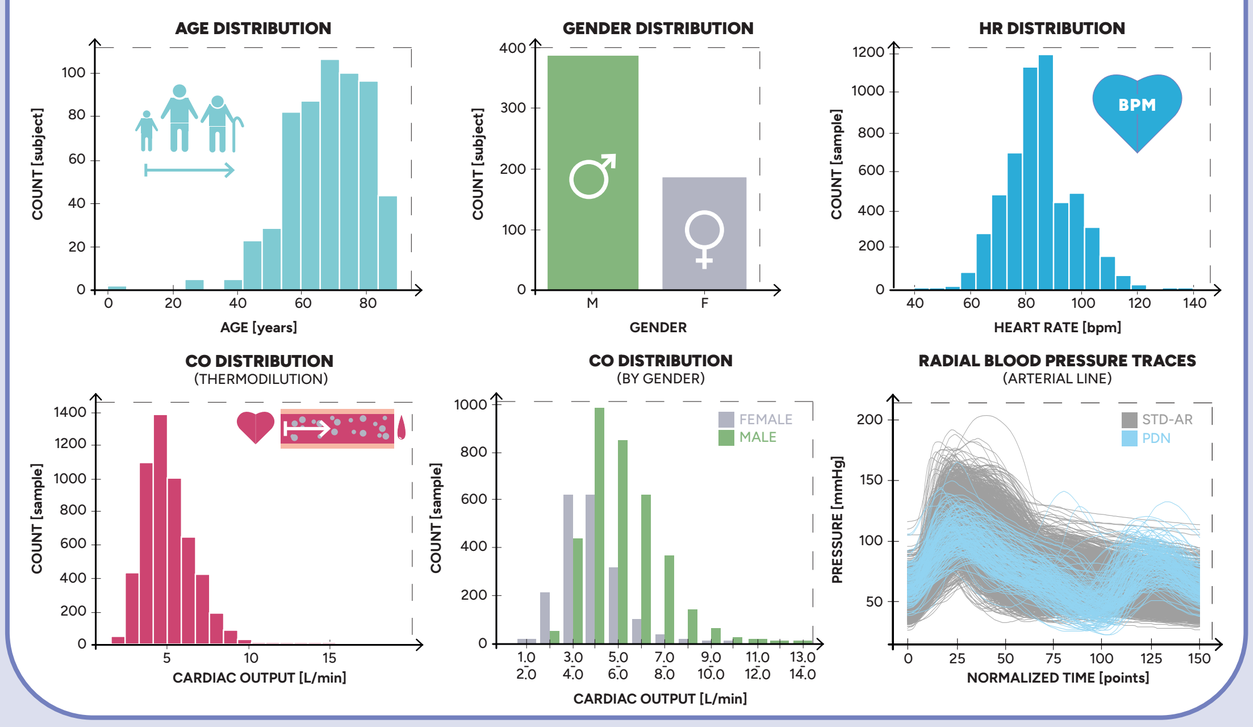
4. Results

Dataset Content

→ 5,454 validated samples from 583 patients
→ Each sample includes CO, HR, ABP, age, gender, MBP, morphology, and SAI score
→ Additional data when available: weight, height, BSA, ICD-9, and ethnicity

Supporting Utilities

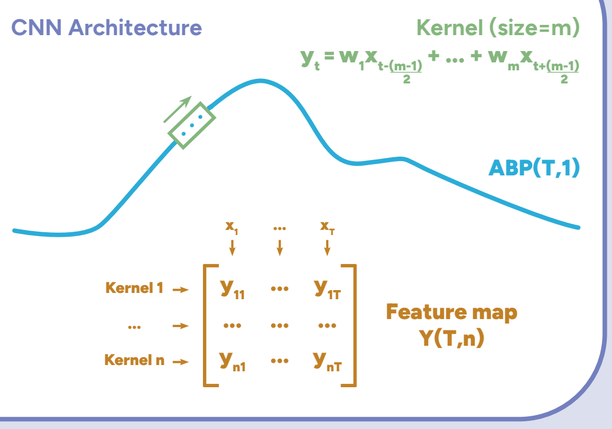
→ A visual review strategy displays all key information on a single plot, enabling fast sample validation
→ A data extraction module generates machine learning–ready datasets automatically



5. Outlook

→ Machine learning to optimize existing CO prediction models (RNN and CNN architectures with multiple inputs: demographic and hemodynamic)
→ Generation of *in silico* pressure wave datasets that better reflect real clinical morphologies, using 1D arterial tree models for synthetic data generation
→ Evaluation of the impact of other CO measurement methods on prediction accuracy (CCO, PiCCO, NICOM, etc.)
→ Integration of photoplethysmography (PPG) signals, widely available in MIMIC-III. The pipeline and processing methods are compatible with PPG

CNN Architecture



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